THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL UNIT TEST SET 2

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. If $a_8 = 25$ and $d = 3$, then $a =$ 1

(a) 4 **(b)** 6 **(c)** 3 **(d)** 5

2. Two poles of heights 13 m and 8 m are 12 m apart. The distance between their tops (m) is: 1

(a) 13 **(b)** 12 **(c)** 15 **(d)** 14

3. If $(1, 2)$, $(3, 6)$ and $(5, k)$ are collinear, then $k =$ 1

(a) 10 **(b)** 9 **(c)** 11 **(d)** 12

4. The 9th term of the AP whose 1st term is 4 and 4th term is 22 is: 1

(a) 51 **(b)** 53 **(c)** 54 **(d)** 52

5. If $DE \parallel BC$ in $\triangle ABC$, $AB = 7$ cm, $AD = 4$ cm, $AC = 21$ cm, then EC (cm) is: 1

(a) 9 **(b)** 10 **(c)** 8 **(d)** 11

6. The perimeter of the triangle with vertices $(0, 0)$, $(5, 0)$, $(0, 12)$ is: 1

(a) 30 **(b)** 32 **(c)** 31 **(d)** 29

7. The number of terms in the AP 8, 11, \dots, 98 is: 1

(a) 33 **(b)** 32 **(c)** 31 **(d)** 30

8. The areas of two similar triangles are 36 cm² and 441 cm². The ratio of their perimeters is 2: 1

(a) 8 **(b)** 9 **(c)** 7 **(d)** 6

9. The point which divides $(8, 0)$ and $(0, 4)$ in the ratio 1:1 has coordinates summing to: 1

(a) 6 **(b)** 7 **(c)** 8 **(d)** 5

10. The sum of all two-digit multiples of 4 is: 1

(a) 1189 **(b)** 1190 **(c)** 1187 **(d)** 1188

11. The altitude to the hypotenuse of a right triangle divides it into 3 cm and 12 cm. The altitude (cm) is: 1

(a) 8 **(b)** 6 **(c)** 5 **(d)** 7

12. The centroid of the triangle with vertices $(0, 0)$, $(6, 3)$, $(3, 9)$ has x-coordinate: 1

(a) 3 **(b)** 5 **(c)** 2 **(d)** 4

13. The sum of the first 20 terms of the AP $-6, -2, 2, 6, \ldots$ is: 1

(a) 642 **(b)** 640 **(c)** 639 **(d)** 641

14. A pole 8 m high casts a shadow 5 m long. A tower casting a shadow 40 m has height (m): 1

(a) 66 **(b)** 65 **(c)** 64 **(d)** 63

15. A circle has centre $(0, 0)$ and passes through $(5, 12)$. Its radius is: 1

(a) 14 **(b)** 15 **(c)** 12 **(d)** 13

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Three vertices of a parallelogram $ABCD$ taken in order are $A(3, 1)$, $B(6, 2)$ and $C(8, 6)$. Find the fourth vertex D. 2

17. Find the sum of all natural numbers between 1 and 100 that are divisible by 4. 2

18. In $\triangle ABC$, D and E are the midpoints of AB and AC and $BC = 8$ cm. Find DE and the ratio of areas of $\triangle ADE$ and $\triangle ABC$. 2

19. Find the centroid of the triangle with vertices $(3, 7)$, $(9, 1)$ and $(6, 4)$. 2

20. Which term of the AP $11, 15, 19, 23, \ldots$ is 79? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. In a right triangle, the altitude to the hypotenuse divides it into segments of 4 cm and 9 cm. Find the altitude. 3

22. Find the area of the triangle with vertices $(2, 5)$, $(6, 1)$ and $(8, 7)$. 3

23. In an AP, the 2nd term is 2 and the 7th term is 32. Find the AP and its 12th term. 3

24. In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AB = 9$ cm, $AD = 2$ cm and $AC = 27$ cm, find AE and EC. 3

OR

Show that the points $(1, 1)$, $(5, 1)$ and $(3, 5)$ are the vertices of an isosceles triangle. 3

25. Find the sum of the first 15 terms of the AP whose nth term is $a_n = 5n - 1$. 3

OR

$\triangle ABC \sim \triangle PQR$ with $AB = 5$ cm, $BC = 6$ cm, $CA = 7$ cm and $PQ = 15$ cm. Find QR and RP. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. The diagonals of a rhombus are 12 cm and 16 cm. Find its side. 5

OR

Find the point on the x-axis which is equidistant from $(1, 5)$ and $(7, 3)$. 5

27. How many terms of the AP $12, 18, 24, 30, \ldots$ must be taken so that the sum is 714? 5

OR

Two poles of heights 20 m and 8 m stand on level ground 16 m apart. Find the distance between their tops. 5

END OF QUESTION PAPER — **Total: 50 Marks**